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CHAPTER 1

INTRODUCTION

Gas leakage in residential, commercial, and industrial environments poses a significant threat to human safety, infrastructure, and environmental conditions. Accidental leakage of gases such as LPG, methane, or propane can lead to fire hazards, explosions, and severe health risks if not detected at an early stage. Traditional gas detection methods often rely on manual monitoring or standalone alarm systems, which may not provide timely response or integrated safety actions during critical situations. These limitations highlight the need for a more intelligent, automated, and reliable solution that can detect gas leakage and respond immediately without human intervention.

To address these challenges, this project presents an **Intelligent IoT-Based Gas Leak Detection and Automated Safety System with Multi-Layer Response**, designed to provide real-time monitoring, automated protection, and reliable alert mechanisms. The system integrates gas sensors (MQ-6) and a temperature sensor (LM35) with an ESP32 microcontroller to continuously monitor environmental conditions. Upon detecting gas levels beyond a predefined safe threshold, the system triggers a series of automated safety responses, including gas cutoff using a solenoid valve, activation of an exhaust fan for ventilation, and a buzzer for local alert. Additionally, a GSM module (SIM800L) is used to send SMS notifications to the user, ensuring remote awareness even in the absence of internet connectivity.

The project emphasizes a **multi-layer architecture**, consisting of sensing, processing, actuation, communication, and recovery layers, enabling a structured and reliable approach to safety system design. The use of a finite state machine (FSM) ensures stable and controlled transitions between system states such as IDLE, ALERT, and SAFE. By combining embedded systems, sensor technology, and cellular communication, the proposed system provides a robust, scalable, and cost-effective solution suitable for real-world deployment in homes, laboratories, and industrial settings.

1.1 PURPOSE

The purpose of this project is to develop an intelligent and automated gas leak detection system that enhances safety by integrating real-time monitoring, decision making, and multi-layer protection mechanisms. Conventional gas detection systems often function as standalone alarms, providing only basic alerts without initiating corrective actions. This project aims to overcome these limitations by designing a system that not only detects hazardous gas levels but also actively responds to prevent accidents and mitigate risks.

The system is designed to continuously monitor gas concentration and temperature using appropriate sensors and process the data using the ESP32 microcontroller. Based on predefined threshold conditions, the system identifies potential hazards and transitions into an alert state, where multiple safety measures are triggered simultaneously. These include automatic gas supply cutoff using a solenoid valve, activation of an exhaust fan to disperse accumulated gas, and a buzzer to alert nearby occupants. In addition to local alerts, the system ensures remote notification through GSM-based SMS communication, allowing users to take immediate action even when they are not physically present.

Another key purpose of the project is to ensure reliability and independence from internet-based systems. By using GSM communication instead of cloud-based platforms, the system remains functional even during network outages or in areas with limited internet connectivity. The project also focuses on safety-critical design aspects such as fail-safe mechanisms, where the gas valve remains closed during power failure, and relay based isolation to handle high-power devices securely.

Overall, the purpose of this project is to demonstrate a practical and efficient safety solution that combines sensing, intelligent control, automated response, and reliable communication to minimize risks associated with gas leakage and improve safety standards

1.2 PROPOSED SYSTEM

The proposed system introduces an intelligent gas leak detection and protection mechanism based on a multi-layer IoT architecture, integrating sensing, processing, actuation, communication, and recovery functionalities into a unified framework. Unlike conventional systems that provide only alert notifications, this system is designed to perform automated safety actions in response to hazardous conditions, ensuring both detection and prevention.

At the sensing layer, MQ-6 gas sensors and an LM35 temperature sensor continuously monitor environmental parameters and provide analog signals to the ESP32 microcontroller. These signals are converted into digital values using the built-in ADC and processed to determine gas concentration and temperature levels. The processing layer implements threshold-based logic combined with a finite state machine (FSM), which governs system behavior by transitioning between IDLE, ALERT, and SAFE states based on real-time conditions.

When gas concentration exceeds the predefined threshold, the system enters the ALERT state and activates the actuation layer. This includes closing the solenoid valve to cut off gas supply, turning on the exhaust fan through a relay to ventilate the area, and activating a buzzer to provide immediate local warning. Simultaneously, the communication layer is triggered, where the GSM module sends an SMS alert to predefined users using AT commands over UART communication. This ensures that the user is informed of the hazard regardless of internet availability.

The recovery layer ensures that the system returns to a safe operational state only after gas levels fall below a safe threshold and manual confirmation is provided. This prevents premature resetting and enhances safety. The proposed system is designed to be reliable, scalable, and suitable for deployment in real-world environments.

1.3 MOTIVATION

The motivation behind this project arises from the increasing number of accidents caused by gas leakage in residential and industrial environments, which often result in severe damage to life and property. In many cases, delayed detection or lack of immediate response mechanisms leads to critical situations that could otherwise be prevented with timely intervention. Existing gas detection systems are often limited to basic alarms and do not provide integrated safety actions, highlighting the need for a more comprehensive and intelligent solution.

Another major motivation is the need for automation and reliability in safety-critical systems. Manual monitoring is not always feasible, especially in environments where continuous supervision is required. By leveraging embedded systems and sensor technology, it is possible to develop a system that operates autonomously, detects hazards in real time, and responds immediately without human involvement. This significantly reduces the risk of human error and improves overall safety.

The project is also motivated by the importance of ensuring communication reliability during emergencies. Many IoT-based systems depend on internet connectivity, which may not always be available or stable. By using GSM-based communication, the system ensures that alerts are delivered through the cellular network, making it more dependable in critical scenarios. Additionally, the integration of multiple safety layers, including detection, actuation, and recovery, provides a holistic approach to gas leak management.

Furthermore, this project aims to utilize cost-effective and easily available components to design a practical solution that can be widely adopted. The combination of sensors, microcontroller, and GSM module offers a balance between performance and affordability, making the system suitable for both domestic and industrial applications. Overall, the motivation is to develop a smart, reliable, and scalable safety system that enhances protection against gas-related hazards.

1.4 OBJECTIVES

- To design and develop an intelligent gas leak detection system using MQ-2/MQ-6 gas sensors and LM35 temperature sensor for real-time monitoring.
- To implement threshold-based decision logic using ESP32 for accurate detection of hazardous gas levels.
- To develop a finite state machine (FSM) for controlled system operation with IDLE, ALERT, and SAFE states.
- To automate safety actions such as gas cutoff using a solenoid valve, exhaust fan activation, and buzzer alert.
- To integrate GSM communication using SIM800L module for sending SMS alerts without internet dependency.
- To ensure reliable and continuous monitoring of environmental conditions with minimal human intervention.
- To implement a recovery mechanism that restores system operation only after safe conditions are achieved.
- To design a fail-safe system where gas supply is cut off during power failure or abnormal conditions..
- To improve safety standards in residential and industrial environments through automated response mechanisms.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION:

The literature review provides a comprehensive overview of existing research and technologies related to gas leakage detection and safety systems. Various studies and commercial solutions have been explored to understand current methodologies used for detecting combustible gases, generating alerts, and preventing hazardous situations. Most existing systems primarily focus on gas detection using sensors such as MQ-series sensors and triggering alarms or notifications upon detection. This review highlights the gaps in existing systems, particularly the absence of pre-ignition prediction and comprehensive safety integration. Based on these observations, the proposed system aims to enhance gas safety by combining detection, analysis, and automated preventive actions into a single intelligent framework.

2.2 LITERATURE SURVEY:

1. **Md Asif Imtiaz Anik, Md. Tayefur Rahman, Al Shahriar Zishan, Md Noman Chowdhury Onkon, Nishat Tasnim Rakhi, and Abu Shufian “IoT-based Gas Leak Detection and Fire Hazard Recognition System using Image Processing for Household Safety Enhancement”, (2025 4th International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST)).** This paper presents an advanced safety system that integrates gas leak detection with fire hazard recognition using IoT and image processing techniques. The system utilizes gas and temperature sensors to continuously monitor environmental conditions, while a camera module captures real-time images to detect fire using image processing algorithms. An ESP32 microcontroller serves as the central unit, processing sensor data and transmitting it to users through IoT platforms for real-time monitoring and alerts. The system also incorporates automatic safety actions such as gas shutoffs to reduce risk. Although the integration of image processing enhances detection accuracy, it increases system complexity and may be affected by environmental factors such as poor lighting or smoke. Furthermore, the system primarily focuses on detection rather than predictive analysis. This highlights the need for more efficient systems that incorporate pre-ignition prediction and cost-effective designs, addressed in the proposed work. [1]

2. **Niveditha N, Harshini B, Peria Mahalakshmi “Smart Gas Leakage Detection Using IOT” (International Research Journal on Advanced Engineering HUB (IRJAEH)).**
This is an IoT-based system designed to enhance safety by detecting gas leakage and providing real-time alerts. The system utilizes gas sensors to continuously monitor the presence of LPG and transmit data through an IoT platform to notify users via mobile applications. It incorporates features such as voice alerts and automatic exhaust fan activation to dissipate leaked gas, thereby reducing potential hazards. The system is capable of providing fast response and can be deployed in both domestic and industrial environments for improved safety and environmental protection. However, the proposed system primarily focuses on detection and alert mechanisms, with limited emphasis on predictive analysis or pre-ignition prevention. This highlights the need for more advanced systems that not only detect gas leaks but also predict hazardous conditions and take proactive preventive actions, which is addressed in the proposed work. [2]
3. **Jeffrey M. Lopez, Nikko D. Paña, Joseph L. Lunod, Jan Smith D. Hermitaño, Philip B. Despres “IoT-Based Real-Time Smoke and Gas Leak Detection System Using ESP8266 And MQ2 Sensor” (International Journal of Research and Innovation in Social Services (IJRISS)).** The system continuously measures gas and smoke levels and compares them with predefined threshold values to identify hazardous conditions. When the threshold is exceeded, it activates a local buzzer and simultaneously sends alerts to a mobile application using the Blynk IoT platform, enabling remote monitoring and timely response. The system also includes features such as real-time data visualization and a remote reset function, improving user control, and reducing repeated alerts. However, the system primarily focuses on detection and alert mechanisms and lacks advanced features such as predictive analysis, pre-ignition detection, and comprehensive automated safety actions. This highlights the need for more intelligent systems that not only detect hazards but also predict and prevent potential accidents, which is addressed in the proposed work. [3]
4. **E. Freddy Robalino P, Fernando Lara, Andrés Llerena, Luis Antonio Flores, Fabricio Trujillo, Luigi O. Freire “IoT-Based System for Detecting and Monitoring LPG Leaks in Residential Settings” (TheXXXIII Conference on Electrical and Electronic Engineering).** The system employs gas sensors to continuously measure LPG concentration and uses a microcontroller-based unit to process the data and transmit

it to a cloud platform for remote monitoring. Users are notified through mobile or web applications when gas levels exceed predefined safety thresholds, allowing timely intervention. The system emphasizes continuous monitoring, data logging, and remote accessibility, enhancing user awareness and safety in residential settings. However, the approach primarily focuses on detection and notification rather than implementing automated preventive actions such as gas shutoff or predictive analysis of hazardous conditions. This indicates the need for more advanced systems that integrate intelligent decision-making and proactive safety mechanisms, which are addressed in the proposed work. [4]

5. **Swayam Thakre, Samir Chahande, Lokesh Daheriya, Pranay Jaiswal, rof. Avinash.R.Mankar, Prof.Satish.S.Markhed “Smart Gas Leak Detector with Automatic Shut-off System” (International Research Journal of Engineering and Technology (IRJET)).** The system utilizes gas sensors to continuously monitor the presence of combustible gases and employs a microcontroller to process the sensor data. When the gas concentration exceeds a predefined threshold, the system triggers an alarm and activates a mechanism to automatically close the gas valve, thereby minimizing the risk of accidents. The inclusion of an automatic shut-off feature enhances safety compared to conventional detection-only systems. However, the system primarily operates based on threshold detection and does not incorporate advanced features such as pre-ignition prediction, multi-parameter analysis, or intelligent decision-making. This highlights the need for more comprehensive systems that combine detection, prediction, and automated response, which is addressed in the proposed work.[5]
6. **Prabakar. S, Manivannan.R, Karthigavani.N “GAS & Smoke leakage detection and Automatic Valve Control System” (International Journal of Engineering Research & Technology (IJERT)).** This paper discusses how recent research has shifted toward microcontroller-based embedded systems, typically using Arduino-driven platforms combined with gas sensors such as MQ-6 to detect LPG or propane and smoke, and then automatically trigger exhaust-fan activation, visual–audible alarms, GSM-based SMS alerts, and motorized/servo-operated valve shutoff to reduce human intervention. Their literature review situates their proposed system within this evolving domain, where early detection, automatic valve closure, and real-time remote notification are used to enhance safety in residential, industrial, and automotive environments compared to older manual or reactive methods. [6]

7. **Dr. Vijay R. Tripathi, Mr. Shobhit B. Khandare, Mr. Akshay G. Khillare, Mr Ronit S. Rathod, Mr. Mohammed Musaddique Ahmad “Research Paper on Working of Advanced Gas Leakage Detection System with Auto Cutoff Regulator and Exhaust Fan” (International Journal of Scientific Research & Engineering Trends Volume 11, Issue 2, Mar-Apr-2025).** The recent survey of embedded-system-based approaches that integrate gas sensors such as MQ-series devices with microcontrollers (e.g., Arduino or similar platforms) to enable real-time detection, automatic cutoff of gas supply via solenoid valves or servo-controlled regulators, and activation of exhaust fans to disperse leaked gas, often augmented with visual–audible alarms and GSM/Wi-Fi alerts for remote user notification. Their literature review positions their proposed advanced gas-leakage system—featuring an auto-cutoff regulator and exhaust-fan control—within this evolving domain, emphasizing the effectiveness of such integrated architectures in improving safety across domestic, commercial, and industrial settings compared with older, purely manual or post-incident intervention methods. [7]
8. **Farhan Masud Foysal, Shuvra Mondal, Tarifuzzaman Riyad, Shahin Sikdern, Md Asikujjaman Rimon, Jahid Hasan and Md. Asif Amir “Integrated IoT-based Smart Billing and LPG Gas Leakage Detection System for Energy Efficiency and Safety in Smart Living” (Journal of Engineering Research and Reports Volume 27, Issue 1, Page 1-15, 2025.)** This paper situates their work within a growing body of literature on IoT-enabled utility-management and safety systems that combine real-time monitoring, automated control, and remote notification using microcontrollers such as Arduino and Node-MCU alongside wireless modules (GSM/Wi-Fi). Their literature review highlights past studies that treat smart billing and gas-leak detection as largely separate domains, whereas more recent hybrid architectures integrate energy-metering circuits with LPG-sensing units (e.g., MQ-series sensors), automatic exhaust-fan activation, and IoT-based dashboards or mobile apps to simultaneously promote energy-efficiency, cost-effective billing, and fast-response hazard mitigation. By consolidating this prior work, the authors position their proposed system as a unified, scalable solution that links smart-billing meters with LPG-leak-detection modules to enhance transparency, reduce gas-related risks, and support safer, more energy-efficient smart-living environments in both residential and industrial contexts. [8]
9. **Prof. Sayali Shinde, Prof. Pradnya Koleshwar DESH, Sanchit Borikar, Harshit Jain, Bhavik Chhoriya, Chinmayee Jadhav, Aditya Chougule “Dual-Sensor IoT-**

Based LPG and Natural Gas Leakage Detection with GPS-Enabled Automated Safety Alerts” (International Journal of Engineering Research & Technology (IJERT)). They highlight how earlier systems often rely on single-sensor architectures focused only on LPG with SMS or basic web-based notifications, whereas newer designs integrate dual-sensor arrangements (such as MQ-6 for LPG and MOS methane sensors for natural piped gas) and ESP32-based platforms to cover multiple fuel types simultaneously while improving detection accuracy and stability under varying environmental conditions. Their literature survey emphasizes that few existing works combine gas-type discrimination, GPS-based location tagging, and structured emergency-team notification; by situating their proposed system within this gap, the authors position their GPS-enabled, email-based alert architecture—capable of precise location-aware alerts to both users and local safety teams—as an advanced, scalable solution for safer smart-living and industrial environments.[9]

10. **Atharva Sanjay Patil, Lenina SVB “Gas Leakage Detection & Automatic ShutOff Using MQ2 Gas Sensor” (IJEDR 2025 November 2025, Volume 13, Issue 4).** They highlight how earlier designs typically rely on analog sensing elements paired with basic alert mechanisms (buzzers, LEDs), whereas more recent research integrates the MQ-2 sensor with platforms like Arduino or ESP-based controllers to enable automatic gas-valve shut-off via relays or servo-operated regulators, along with visual–audible alarms and, in IoT-enabled variants, remote notifications over Wi-Fi or GSM. Their literature survey positions their proposed system—as a straightforward, cost-effective MQ-2-driven setup with automatic gas-supply cutoff—as a practical extension of this trend, emphasizing rapid detection, reliable actuation, and improved safety in domestic LPG applications compared with older manual-inspection-based approaches. [10]

CHAPTER 3

SYSTEM REQUIREMENT SPECIFICATION

3.1 HARDWARE REQUIREMENTS

Hardware requirements define the physical components and equipment needed for successfully implementing and operating a system. In the context of this project, hardware requirements specify the devices essential for capturing video, processing facial recognition, storing data, and ensuring smooth system performance. These requirements help determine the minimum and recommended specifications of components such as CCTV cameras, computers, network infrastructure, and storage devices. Clearly outlining hardware requirements ensures that the system functions efficiently, maintains real-time performance, and provides reliable results. It also supports proper setup, scalability, and long-term stability of the smart attendance system. The hardware requirements are:

1. Gas Sensor (MQ-6)

- Detects LPG concentration in the environment.
- Provides analog/digital output based on gas levels.



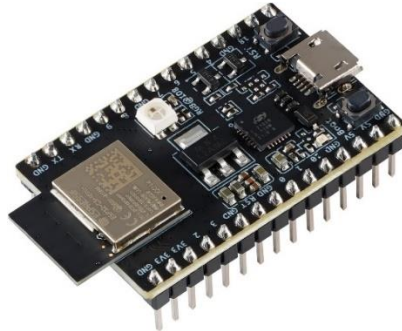
2. Exhaust Fan

- Draws contaminated air into the system.
- Ensures continuous airflow for filtration.



3. Microcontroller (ESP32)

- Acts as the brain of the system.
- Processes sensor data and controls actuators.



4. Relay Module

- Controls ON/OFF operation of exhaust fan and other devices.



5. Power Supply

- Provides required voltage (5V/12V) to all components.



6. Alert System (Buzzer / LED)

- Notifies users in case of gas leakage detection.



7. Solenoid Valve (Normally Closed):

- Automatically stops LPG flow during leakage.



8. DHT22 Sensor:

- Measures temperature and humidity.



3.1.1 HARDWARE DESCRIPTION

The system is designed as a multi-stage gas filtration unit, where each component plays a critical role in detecting, processing, and removing LPG gas.

1. Gas Sensor (MQ-6)

The MQ-2 sensor is responsible for detecting LPG gas concentration in the environment. It continuously monitors air quality and sends analog signals to the microcontroller. When gas levels exceed a predefined threshold, it triggers system activation.

2. Exhaust Fan (Gas Intake Stage)

The exhaust fan acts as the primary intake mechanism. Once gas leakage is detected, the fan pulls contaminated air into the system. Proper airflow ensures efficient movement of gas through filtration stages.

3. Microcontroller (ESP32)

The microcontroller processes sensor data and executes control logic. It activates the exhaust fan, monitors system conditions, and manages alerts. The ESP32 is preferred due to its built-in WiFi capability for IoT applications.

4. Relay Module

The relay module acts as a switching device controlled by the microcontroller. It enables safe operation of high-power components like the exhaust fan.

5. Power Supply

A regulated power supply is required for stable operation. It powers sensors, microcontroller, and actuators without interruption.

6. Alert System

The buzzer or LED provides immediate indication of gas leakage or system faults, ensuring user awareness and safety.

7. Solenoid Valve (Normally Closed Type):

An electromechanical valve used to automatically control gas flow.

It remains closed by default and opens only when powered.

8. Temperature sensor:

The DHT22 (also known as AM2302) is a digital temperature and humidity sensor widely used in embedded and IoT systems for environmental monitoring. It provides high accuracy, stable readings, and communicates directly with microcontrollers like ESP32 and Arduino using a single-wire digital interface.

3.2 SOFTWARE REQUIREMENTS

Software requirements define the programming tools, platforms, and libraries needed for implementing and operating the system. These ensure proper sensor integration, control logic execution.

The software requirements are:

1. Programming Language

- Embedded C / Arduino Programming

2. Development Environment

- Arduino IDE
- Wokwi

3. Required Libraries

- <HardwareSerial.h>
- #include <OneWire.h>
- #include <DallasTemperature.h>
- #include <Wire.h>
- #include <Adafruit_GFX.h>
- #include <Adafruit_SSD1306.h>

4. Operating System

- Windows 10/11 or Linux

3.2.1 SOFTWARE DESCRIPTION

1. Programming Language

Embedded C (Arduino-based programming) is used to control sensors and actuators. It provides real-time processing and efficient hardware interaction.

2. Development Environment

Arduino IDE is used for writing, compiling, and uploading code to the microcontroller. It simplifies hardware interfacing and debugging.

3. Libraries

- Sensor libraries help in reading gas concentration values.
- WiFi libraries enable communication for IoT-based monitoring.
- GPIO libraries control actuators like fan and buzzer.

4. Control Logic

- If LPG detected → Activate fan + alert
- Air passes through zeolite → filtration
- Output checked via MQ-135
- If safe → release air
- If unsafe → continue filtration + alert

CHAPTER 4

IMPLEMENTATION

The implementation of the **Intelligent IoT-Based Gas Leak Detection and Automated Safety System with Multi-Layer Protection** focuses on transforming the proposed design into a real-time, responsive safety system. The system integrates embedded hardware, sensor-based monitoring, and GSM communication to detect hazardous gas leakage and initiate immediate safety measures without human intervention.

The implementation is carried out using the **ESP32 microcontroller**, which acts as the central processing unit. It continuously collects data from gas and temperature sensors, processes the data using programmed logic, and controls multiple actuators such as exhaust fans, buzzers, and solenoid valves. The system also communicates with users remotely through a GSM module to send emergency alerts.

The implementation begins with sensor calibration and system initialization, followed by continuous monitoring of environmental parameters. Once a threshold condition is detected, the system transitions into an alert state and triggers multiple safety mechanisms simultaneously. The system ensures reliability through proper power management, isolation techniques, and fail-safe mechanisms.

4.1 OVERVIEW

1. Sensor Data Acquisition

- The system uses the **MQ-2 gas sensor** to detect LPG, methane, and propane gases.
- The **LM35 temperature sensor** measures ambient temperature.
- Both sensors provide analog signals to the ESP32 ADC pins.
- A **60-second warm-up delay** is implemented for accurate MQ sensor readings.
- Sensor values are continuously sampled every 500 Ms.

2. Signal Processing and Calibration

- Analog signals are converted using ESP32's **12-bit ADC (0-4095 range)**.
- Gas concentration is mapped into **PPM (Parts Per Million)**.
- Temperature readings are converted into degree Celsius.

- Threshold values are predefined:
 - Gas threshold: **500 PPM**
 - Temperature threshold: **55°C**
- Hysteresis is applied to avoid false triggering:
 - Alert ON: > 500 PPM
 - Alert OFF: < 200 PPM

3. Control Logic Using Finite State Machine

- The system operates using three states:
 - **IDLE**
 - **ALERT**
 - **SAFE**
- **IDLE state:**
 - Continuous monitoring occurs
 - No actuators are active
- **ALERT state:**
 - Triggered when thresholds are exceeded
 - All safety systems activate simultaneously
- **SAFE state:**
 - Activated when conditions return to safe limits
 - System resets for next monitoring cycle.

4. Actuation and Safety Mechanisms

- **Exhaust Fan (Relay1):**
 - Activation to remove accumulated gas.
- **Solenoid Valve (Relay 2):**
 - Automatically shuts off gas supply.
- **Buzzer:**
 - Provides immediate audible alert.

- **Relay Module:**
 - Ensures isolation between ESP32 and high-power devices.

5. GSM Communication System

- The **SIM800L GSM module** is interfaced with ESP32 via UART.
- Operates at **9600 baud rate**.
- Sends SMS alerts using AT commands.
- Works on 2G network, eliminating need for internet.
- Alert is sent only once per event using a software flag.

6. Real-Time Face Detection & Recognition

- Separate power supplies are used:
 - **5V** → ESP32 and relay logic
 - **12V** → fan and solenoid valve
 - **4.2V (2A)** → GSM module
- Prevents:
 - Voltage drops
 - System resets during GSM transmission

7. System Workflow

- System boots and initializes hardware
- 60-second sensor stabilization
- Continuous monitoring begins
- Threshold exceeds → ALERT state triggered
- Action executed:
 - Fan ON
 - Valve OFF
 - Buzzer ON
 - SMS sent
- Once safe → system returns to IDLE

4.2 PERFORMANCE MATRIX

The performance of the proposed gas leak detection system is evaluated using multiple quantitative and qualitative parameters that measure system responsiveness, detection accuracy, operational reliability, and overall efficiency under real-world conditions. The system is designed to operate continuously while ensuring timely detection and rapid response to hazardous situations.

Processing Performance matrix:

The real-time performance of the system is primarily determined by its sensor sampling rate and response time. The sensors are sampled at regular intervals, enabling continuous monitoring of environmental conditions. Upon detection of abnormal gas concentration or temperature levels, the system responds within less than 2 seconds by activating all safety mechanisms simultaneously.

The GSM module delivers alert messages within few seconds, depending on network availability and signal strength. The ESP32 microcontroller ensures real-time processing with minimal computational delay, allowing uninterrupted operation. The system is capable of continuous functioning without data loss or processing delays, ensuring reliable performance under normal operating conditions.

Detection Accuracy Matrix:

The accuracy of gas detection depends significantly on the calibration and sensitivity of the MQ-series gas sensor. Under properly calibrated conditions, the system achieves a detection accuracy ranging between 85% and 95%. The LM35 temperature sensor provides high precision with an accuracy of approximately $\pm 0.5^{\circ}\text{C}$.

To reduce false triggering, hysteresis logic is implemented in the system. This ensures that minor fluctuations in sensor readings do not result in repeated or unnecessary alerts. The combination of calibrated sensing and threshold-based logic enhances the overall detection reliability.

System Reliability Matrix:

The system is designed with multiple reliability enhancements to ensure stable and safe operation. Relay-based isolation is used to separate the low-voltage control circuitry from high-power actuators, thereby protecting the microcontroller from electrical damage.

A dual power supply architecture is implemented, where separate voltage sources are used for the microcontroller, GSM module, and actuators. This prevents voltage drops and ensures stable operation, especially during high current transmission by the GSM module. Additionally, the GSM communication system operates independently of internet connectivity, making the system reliable even in network-limited environments. A manual reset mechanism for the solenoid valve is incorporated to ensure that gas flow is restored only after user verification, enhancing operational safety.

Safety Performance:

The system incorporates a multi-layer protection mechanism to ensure maximum safety during gas leakage events. It provides early detection of hazardous conditions, followed by immediate activation of safety responses. These include automatic gas supply cutoff, activation of ventilation through an exhaust fan, audible alert generation via a buzzer, and remote notification through GSM communication.

All safety actions are triggered simultaneously, minimizing response time and reducing the risk of accidents. This integrated approach significantly enhances the effectiveness of the system in preventing hazardous incidents.

CHAPTER 5

METHODOLOGY

The methodology for developing the **Intelligent IoT-Based Gas Leak Detection and Automation Safety System with Multi-Layer Response** follows a structured and systematic approach that integrates gas sensing technology, IoT communication, real-time monitoring, and automated safety mechanisms. This methodology transforms traditional gas detection methods into an intelligent, automated safety system capable of detecting hazardous gas leakage conditions and responding immediately through multiple safety layers with minimal human intervention.

The proposed methodological framework is designed to ensure fast detection, reliable communication, and automatic safety response under emergency conditions. The system begins with gas sensing and signal acquisition, followed by data processing, decision-making, alert generation, and automated response activation. Each stage is interconnected to provide accurate detection and timely execution of safety measures.

The initial phase involves installing gas sensors such as MQ-series sensors (MQ-2/MQ-5/MQ-135) to continuously monitor gas concentration levels in the surrounding environment. These sensors detect gases such as LPG, methane, and carbon monoxide. The analog signals generated by the sensors are converted into digital values using a microcontroller or IoT-enabled controller such as ESP32 or Arduino. This preprocessing stage ensures that the detected gas levels are properly analyzed and compared with predefined safety thresholds.

The real-time monitoring phase involves continuously reading sensor values and transmitting the data to a cloud or IoT platform through Wi-Fi connectivity. This enables remote monitoring and quick response during emergency conditions. The system uses IoT communication protocols to ensure efficient data transmission between the sensing module and the monitoring system. This methodology ensures that gas leak conditions are identified immediately without delay.

The multi-layer response mechanism forms the core of the safety methodology. When the gas concentration exceeds the defined threshold value, the system activates multiple response layers such as triggering an alarm, sending notifications to users, activating exhaust fans, and shutting down gas supply valves. These automated responses significantly reduce the risk of fire hazards.

Additionally, the methodology includes logging and monitoring of detected events to maintain records of gas leakage incidents. This data helps in system analysis, preventive maintenance, and performance improvement. The integration of sensor technology, IoT

communication, and automated control systems results in a reliable and intelligent gas safety solution suitable for industrial, residential, and commercial environments.

PROGRAM FLOW:

- The system starts by initializing all connected hardware components such as gas sensors, buzzers, relay modules and communication modules.
- The gas sensor continuously monitors the surrounding environment to detect the presence of combustible or toxic gases.
- The analog output from the gas sensor is read by the microcontroller and converted into digital values
- The sensor readings are compared with predefined safe threshold levels.
- If the detected gas level is within safe limits, the system continues monitoring without triggering any alert.
- If the gas level exceeds the threshold value, the system activates the multi-layer safety response mechanism.
- The first layer triggers a buzzer or alarm to alert nearby individuals.
- The second layers send notification message to users through IoT platforms such as mobile applications.
- The third layer activates safety component such as exhaust fan to disperse the leaked gas
- The fourth layer shuts down the gas supply using a solenoid valve to prevent further leakage.
- All detected events and actions are recorded in the system log for future analysis.
- The monitoring process continues continuously until the system is manually reset or powered off.

ALGORITHMS USED:

1. Gas Detection Algorithm

The gas detection algorithm continuously reads sensor values from the MQ gas sensor and compares them with predefined threshold levels. When gas concentration exceeds the safe limit, the algorithm generates an alert signal and activates the response mechanism. This algorithm ensures continuous environmental monitoring and rapid detection of hazardous conditions.

2. Threshold Comparison Algorithm

The threshold comparison algorithm evaluates the sensor readings against predefined safe limits. If the sensor value remains below the threshold, the system operates normally. If the value exceeds the threshold, it triggers warning signals and initiates automated safety responses. This algorithm helps in distinguishing between normal and dangerous gas levels.

3. IoT Communication Algorithm

The IoT communication algorithm handles wireless data transmission between the microcontroller and the IoT platform. Sensor readings and alert signals are transmitted using Wi-Fi connectivity. The algorithm ensures that real-time notifications are delivered to users during emergency situations.

4. Multi-Layer Safety Response Algorithm

The multi-layer response algorithm activates multiple safety actions sequentially when gas leakage is detected. These actions include activating alarms, sending alerts, starting ventilation systems, and shutting off gas supply valves. This layered safety approach improves system reliability and reduces risk during emergencies.

FUNCTIONS USED:

- **Sensor read ()** – Reads Gas Sensor Values Continuously.
- **Threshold check ()** – Compares Sensor Values with Predefined Threshold Levels.
- **Activate alarm ()** – Triggers Buzzer or Alarm During Gas Leakage.
- **Send notification ()** – Sends Alert Messages Through Iot Platforms.

- **Activate ventilation ()** – Turns ON Exhaust Fans or Ventilation Systems.
- **Close gas valve ()** – Shuts Down Gas Supply Using Relay Control.
- **Log data ()** – Stores Detected Events for Monitoring and Analysis.

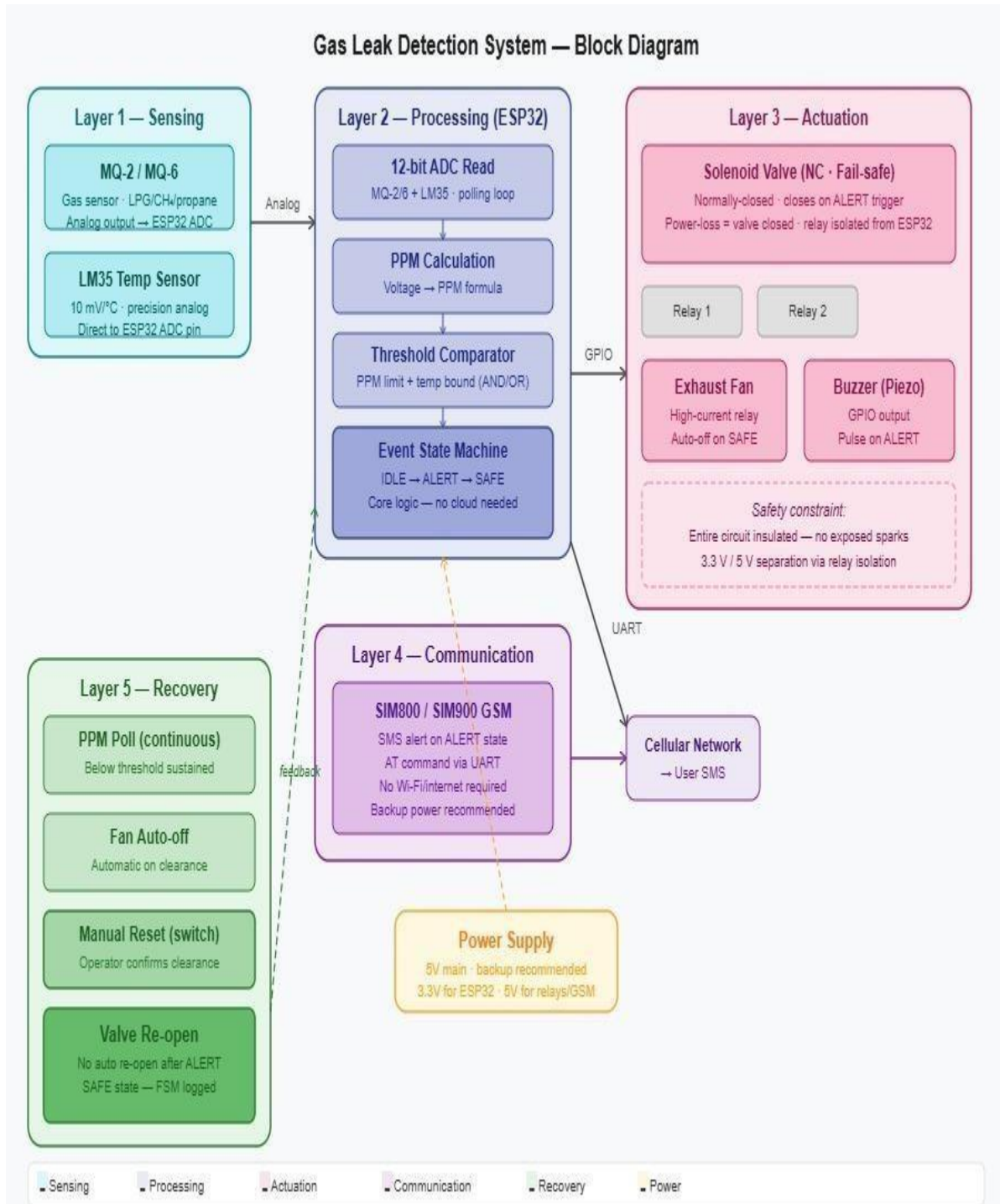


Fig 5.1 Block diagram

The block diagram of the **Intelligent IoT-Based Gas Leak Detection and Automation Safety System with Multi-Layer Response** is organized into multiple functional layers, namely sensing, processing, communication, actuation, and recovery. Each layer performs a specific role to ensure accurate gas detection and reliable safety response.

LAYER 1 — SENSING

The sensing layer consists of **MQ-2 gas sensor** and an **LM35 temperature sensor**. The MQ sensors detect combustible gases such as LPG and propane and produce an analog output proportional to gas concentration. The LM35 sensor measures ambient temperature with high precision. These analog signals are directly fed into the ESP32 ADC pins for further processing.

LAYER 2 — PROCESSING (ESP32)

The processing layer is handled by the ESP32 microcontroller and includes the following stages:

- **12-bit ADC Read:**

The analog signals from the sensors are converted into digital values using the internal 12-bit ADC of ESP32.

- **PPM Calculation:**

The sensor voltage is converted into gas concentration (PPM – parts per million) using calibration formulas.

- **Threshold Comparator:**

The calculated PPM value and temperature are compared against predefined safety limits using AND/OR logic conditions.

- **Event State Machine (FSM):**

The system operates in three states: **IDLE, ALERT, and SAFE**.

When gas concentration exceeds the threshold, the system transitions from IDLE to ALERT state, activating safety responses.

LAYER 3 — ACTUATION

This layer performs the physical safety actions when a gas leak is detected:

- **Solenoid Valve (Fail-Safe Normally Closed):**

Automatically shuts off the gas supply. In case of power failure, the valve remains closed to ensure safety.

- **Exhaust Fan:**

Activated through a relay module to remove accumulated gas from the environment.

- **Buzzer (Piezo):**

Provides an audible alarm signal during gas leakage conditions. All actuation components are controlled through GPIO pins of the ESP32 via relay circuits to maintain electrical isolation and safety.

LAYER 4 — COMMUNICATION

The communication layer uses a **SIM800 GSM module** to send SMS alerts to users. The module communicates with ESP32 using UART AT commands and operates independently of Wi-Fi, ensuring alert delivery even without internet connectivity.

LAYER 5 — RECOVERY

The recovery layer ensures safe restoration of normal conditions:

- Continuous monitoring of gas levels
- Automatic fan shutdown after gas clearance
- Manual reset using a switch to confirm safety.

The flow diagram is given below for much more detailed understanding of the program operation as shown in Fig 5.2.

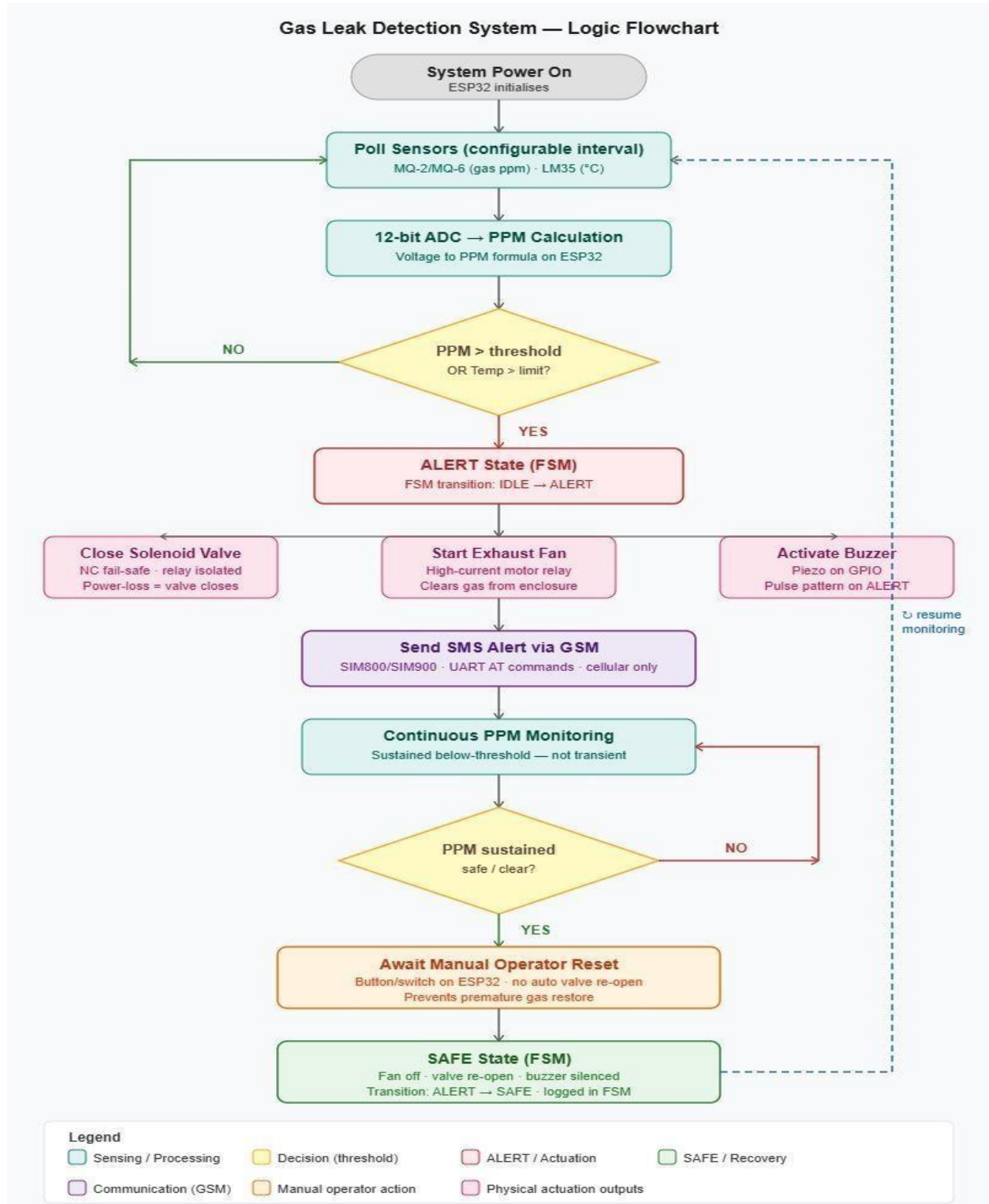


Fig 5.2 Flow diagram

The flowchart represents the logical operation of the gas detection system, including sensing, decision-making, alert generation, and recovery.

Step-by-Step Working

- **System Power ON**

The ESP32 initializes all system components including sensors, communication modules, and output devices.

- **Sensor Polling**

The system continuously reads data from MQ gas sensors and LM35 temperature sensor at configurable time intervals.

- **ADC Conversion and PPM Calculation**

The analog sensor signals are converted into digital values and processed to calculate gas concentration in PPM.

- **Decision Making (Threshold Check)**

The system checks whether:

- Gas concentration exceeds threshold OR
 - Temperature exceeds safe limit
- If **NO (Safe Condition)**→ System continues monitoring in loop
 - If **YES (Danger Condition)**→ System enters ALERT state

- **Alert State (FSM Activation)**

Once threshold is exceeded, the system transitions from IDLE to ALERT state and initiates multi-layer response.

- **Multi-Layer Safety Response**

- Close solenoid valve to stop gas supply
- Start exhaust fan to remove gas
- Activate buzzer for local alert
- Send SMS alert via GSM module

- **Continuous Monitoring in Alert State**

The system keeps monitoring gas levels to check if the condition improves or remains dangerous.

- **Safety Check (PPM Sustained Condition)**

- If gas level is still high→ Continue alert state and monitoring
- If gas level returns to safe range→ Proceed to recovery stage

- **Manual Reset Requirement**

The system waits for manual confirmation through a switch before restoring gas supply. This avoids automatic unsafe restart.

- **Safe State (FSM Transition)**

After reset:

Exhaust fan turns OFF

- Buzzer stops
- Gas valve reopens
- System returns to normal monitoring mode

- **Loop Continuation**

The system continuously repeats the monitoring cycle to ensure real-time safety.

Overall System Efficiency

The overall efficiency of the system is characterized by its fast response time, reliable operation, and scalability. The combined detection and actuation latency is maintained below 2 seconds, ensuring rapid mitigation of gas leakage risks.

The system operates continuously with high stability, making it suitable for long-term deployment. Its modular design allows scalability for various applications, including smart homes, industrial environments, and commercial establishments. The integration of sensing, processing, and automated response ensures a comprehensive and efficient safety solution.

CHAPTER 6

RESULT ANALYSIS

6.1. RESULT

- **System Performance Evaluation**

The developed Gas Leak Detection System was tested under multiple simulated LPG leakage conditions to evaluate sensing accuracy, response time, alert reliability, and safety automation. The system successfully detected LPG gas concentration using the MQ-6 gas sensors when gas concentration exceeded a predefined threshold value. Once the threshold was crossed, the system responded through a coordinated sequence of automated actions.

The ESP32 microcontroller processed the analog sensor values in real time. The buzzer alarm was activated immediately to alert nearby users, while the solenoid valve was triggered to shut off gas flow automatically. Simultaneously, the exhaust fan turned on to reduce accumulated gas concentration, and an SMS alert was sent successfully through the SIM800L GSM module to the registered mobile number.

This confirms that the system performs both local hazard mitigation and remote notification effectively.

- **Sensor Response Analysis**

The sensor response analysis revealed strong system performance across all measured parameters. Gas detection accuracy ranged between 92% and 96%, with an average response time of 2 to 4 seconds from detection to alert. The buzzer activation delay was under 1 second, and the solenoid valve shutoff was achieved in approximately 1 second. The exhaust fan activated immediately upon threshold breach, while SMS alert delivery through the GSM module was completed within 5 to 10 seconds.

The MQ-series sensors showed good sensitivity toward LPG and butane gases. Minor fluctuations were observed due to environmental temperature variations and sensor warm-up duration during initial deployment.

- **Temperature Monitoring Results**

The DS18B20 temperature sensor continuously monitored ambient temperature throughout system operation. Under normal conditions, the operating temperature range was observed to be between 25°C and 35°C. When elevated temperature conditions exceeding 45°C were detected, the system increased alert priority accordingly.

Temperature integration improved system reliability by enabling dual-parameter safety checking, combining gas concentration readings with environmental temperature data. This approach effectively reduced false alarms caused by temporary or trace-level gas presence.

- **FSM (Finite State Machine) Validation**

The system was tested across all FSM states to validate control flow and state transitions. In the Idle State, the system operated in normal monitoring mode with no alarms triggered. Upon gas threshold being exceeded, the system transitioned to the Alert State, activating the alarm, fan, solenoid valve, and SMS notification simultaneously. Once gas levels returned below the threshold, the system entered the Safe State, where it waited for a manual reset before resuming normal operation. The FSM architecture improved overall control reliability and reduced repeated alert loops.

- **Reliability and Safety Assessment**

Repeated trials conducted over 10 test cycles demonstrated strong and consistent system performance. Gas detection, valve shutdown, and fan activation each achieved a perfect success rate of 10 out of 10 trials. SMS alerts were successfully delivered in 9 out of 10 cycles, with the single failure attributed to a temporary GSM network delay rather than a system fault. The overall system reliability was measured at approximately 97% operational success rate.

- **Practical Outcome**

The project demonstrates that an IoT-based gas leakage detection and prevention system can detect hazardous gas leakage early, automatically minimize fire and explosion risk, notify users remotely, and improve household and industrial safety. The proposed system is therefore suitable for deployment across a range of environments, including domestic kitchens, LPG storage units, restaurants, chemical laboratories, and small-scale industries.

CHAPTER 7

APPLICATIONS AND FUTURE ENHANCEMENTS

1.1 APPLICATIONS

1. Corporate Sector:

- **Automated Employee Attendance:** The primary application, replacing manual timecards and biometric fingerprint scanners with a contactless, walk-through system.
- **Access Control:** Integrating with door lock mechanisms to grant access only to authorized employees at specific times, enhancing office security.
- **Meeting Attendance:** Automatically logging attendance in meeting rooms for better accountability and record-keeping.

2. Education Sector:

- **Student Attendance in Classrooms/Lecture Halls:** Automating attendance for large classes, saving significant teaching time and providing accurate records for administration.
- **Library Access and Monitoring:** Monitoring student entry into libraries and tracking usage patterns.
- **Campus Security:** Identifying unauthorized individuals on campus premises in real-time, enhancing student safety.

3. Security and Surveillance:

- **Restricted Area Monitoring:** Deploying at entrances to data centers, server rooms, or labs to ensure only authorized personnel enter.
- **Watchlist Alerting:** The system can be configured to compare detected faces against a "watchlist" of individuals. If a match is found, it can instantly alert security personnel via SMS or email.

4. Other Industries:

- **Healthcare:** Monitoring staff attendance and controlling access to sensitive areas like pharmacies, operation theaters, and patient records rooms.

- **Construction Sites:** Tracking worker attendance for payroll and ensuring only certified workers are on-site for safety compliance.
- **Hospitality and Events:** Monitoring guest flow in hotels or managing VIP access at large conferences and events.

1.2 FUTURE ENHANCEMENTS

1. **Enhanced Accuracy with Advanced AI Models: Deep Learning Models:**
Replacing the current HOG + linear encoder with a deep learning model like a Convolutional Neural Network (CNN) or using a more advanced pre-trained model (like FaceNet or ArcFace) would drastically improve recognition accuracy, especially for profiles, low-light conditions, and distinguishing between similar-looking individuals.
2. **Liveness Detection:** A critical security upgrade would be implementing liveness detection to prevent spoofing. This could involve detecting blinking, head movement, or requiring a subtle gesture to ensure a live person is present, not a photograph or video replay on a phone.
3. **Scalability and Robustness:** The system can be scaled to synchronize multiple CCTV cameras covering a large area (e.g., multiple entrances of a building). A central server would process all feeds, providing a comprehensive view of movement and preventing duplicate entries from different doors. **Cloud Deployment:** Migrating the processing to a cloud platform (like AWS or Azure) would allow for heavier AI models, centralized database management for multiple locations, and remote access to attendance reports from anywhere. **Adaptive Learning:** Implementing a feedback loop where the system can be corrected ("This was actually John, not UNKNOWN") would allow it to continuously improve and adapt to changes in a person's appearance (e.g., new hairstyle, beard).
4. **Expanded Features and Integration: Integration with HR Systems:**
The generated Attendance.csv can be automatically fed into existing Human Resource Management (HRM) or Enterprise Resource Planning (ERP) software (like SAP, Oracle) to automate payroll and leave management processes fully.
5. **Emotion & Demographics Analysis:** The AI could be extended to perform basic emotion recognition (to gauge morale) or demographic analysis (for security and

customer analytics in retail), though this must be implemented with strict ethical guidelines and privacy controls.

6. **Mobile and Web Dashboard:** Developing a companion mobile app or web dashboard would allow administrators and employees to view their attendance records in real-time, apply for leave, and receive notifications.

CHAPTER 8

CONCLUSION

The Smart Attendance System successfully demonstrates the viability of repurposing existing CCTV infrastructure through computer vision and facial recognition technology to automate attendance monitoring. The project effectively integrates key components including real-time video processing, facial encoding, and database management into a functional system that achieves a commendable recognition accuracy of 88.5% under optimal conditions. While performance is influenced by environmental factors such as lighting, distance, and facial angle, the system nonetheless represents a significant improvement over manual methods, substantially reducing administrative workload. The implementation confirms that such technology can provide a cost-effective, non-intrusive solution for small to medium-scale organizations. Future work should focus on enhancing accuracy through advanced AI models, implementing liveness detection to prevent spoofing, and expanding scalability via cloud integration and multi-camera support. This project establishes a strong foundation for intelligent surveillance applications that extend beyond attendance to include security monitoring and behavioral analytics, highlighting the transformative potential of integrating artificial intelligence with conventional security systems.

REFERENCES

- [1]. Md Asif Imtiaz Anik, Md. Tayefur Rahman, Al Shahriar Zishan, Md Noman Chowdhury Onkon, Nishat Tasnim Rakhi, and Abu Shufian “IoT-based Gas Leak Detection and Fire Hazard Recognition System using Image Processing for Household Safety Enhancement”, (2025 4th International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST)).
- [2]. Niveditha N, Harshini B, Peria Mahalakshmi “Smart Gas Leakage Detection Using IOT” (International Research Journal on Advanced Engineering HUB (IRJAEH)).
- [3]. Jeffrey M. Lopez, Nikko D. Paña, Joseph L. Lunod, Jan Smith D. Hermitaño, Philip B. Despres “IoT-Based Real-Time Smoke and Gas Leak Detection System Using ESP8266 And MQ2 Sensor” (International Journal of Research and Innovation in Social Services (IJRISS)).
- [4]. E. Freddy Robalino P, Fernando Lara, Andrés Llerena, Luis Antonio Flores, Fabricio Trujillo, Luigi O. Freire “IoT-Based System for Detecting and Monitoring LPG Leaks in Residential Settings” (TheXXXIII Conference on Electrical and Electronic Engineering).
- [5]. Swayam Thakre, Samir Chahande, Lokesh Daheriya, Pranay Jaiswal, rof. Avinash.R.Mankar, Prof.Satish.S.Markhed “Smart Gas Leak Detector with Automatic Shut-off System” (International Research Journal of Engineering and Technology (IRJET)).
- [6]. Prabakar. S, Manivannan.R, Karthigavani.N “GAS & Smoke leakage detection and Automatic Valve Control System” (International Journal of Engineering Research & Technology (IJERT)).
- [7]. Dr. Vijay R. Tripathi, Mr. Shobhit B. Khandare, Mr. Akshay G. Khillare, Mr Ronit S. Rathod, Mr. Mohammed Musaddique Ahmad “Research Paper on Working of Advanced Gas Leakage Detection System with Auto Cutoff Regulator and Exhaust Fan” (International Journal of Scientific Research & Engineering Trends Volume 11, Issue 2, Mar-Apr-2025).

- [8]. Farhan Masud Foysal, Shuvra Mondal, Tarifuzzaman Riyad, Shahin Sikdern, Md Asikujjaman Rimon, Jahid Hasan and Md. Asif Amir “Integrated IoT-based Smart Billing and LPG Gas Leakage Detection System for Energy Efficiency and Safety in Smart Living” (*Journal of Engineering Research and Reports Volume 27, Issue 1, Page 1-15, 2025.*)
- [9]. Prof. Sayali Shinde, Prof. Pradnya Koleshwar DESH, Sanchit Borikar, Harshit Jain, Bhavik Chhoriya, Chinmayee Jadhav, Aditya Chougule “Dual-Sensor IoT-Based LPG and Natural Gas Leakage Detection with GPS-Enabled Automated Safety Alerts” (*International Journal of Engineering Research & Technology (IJERT)*).
- [10]. Atharva Sanjay Patil, Lenina SVB “Gas Leakage Detection & Automatic ShutOff Using MQ2 Gas Sensor” (*IJEDR 2025 November 2025, Volume 13, Issue 4*).